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# Opening the Black Box: Mechanistic Interpretability of Agent Tool Selection with Sparse Autoencoders

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Interpretability  
for AI Agentssubject Nemotron-3-Nano-30B  
gen·label·judge Qwen3-VL-235B  
compute NVIDIA · 4×GB200

KEY RESULT — CAUSAL, NOT JUST CORRELATED

# 12.8%

of agent tool choices reverse when 10 SAE features are ablated.

Removing a handful of features **causally changes** the agent's tool choice — controls stay flat at zero.random ablation 0% SAE round-trip 0%  $p = 0.002$ also significant: single vs. multi-tool — 17.1% vs. 2.4% baseline ( $n = 41$ ,  $p = 0.029$ ).

shallow vs. deep (tool selection, 1 of 23 contrast types tested) — Nemotron-3-Nano-30B (52 layers), probed at layer 20.

## 01 The Tool-Selection Black Box

Modern AI agents autonomously pick among databases, code, web search, and sub-agents. **When** the agent chooses is observable — **why** it chose remains opaque.

- Prompt engineering — correlational, not causal.
- Behavioural testing — input/output only.
- Chain-of-thought — plausible, often confabulated.

We open the box at the token where the model **commits to a tool**, decomposing its hidden state into human-readable features.

## 02 Pipeline

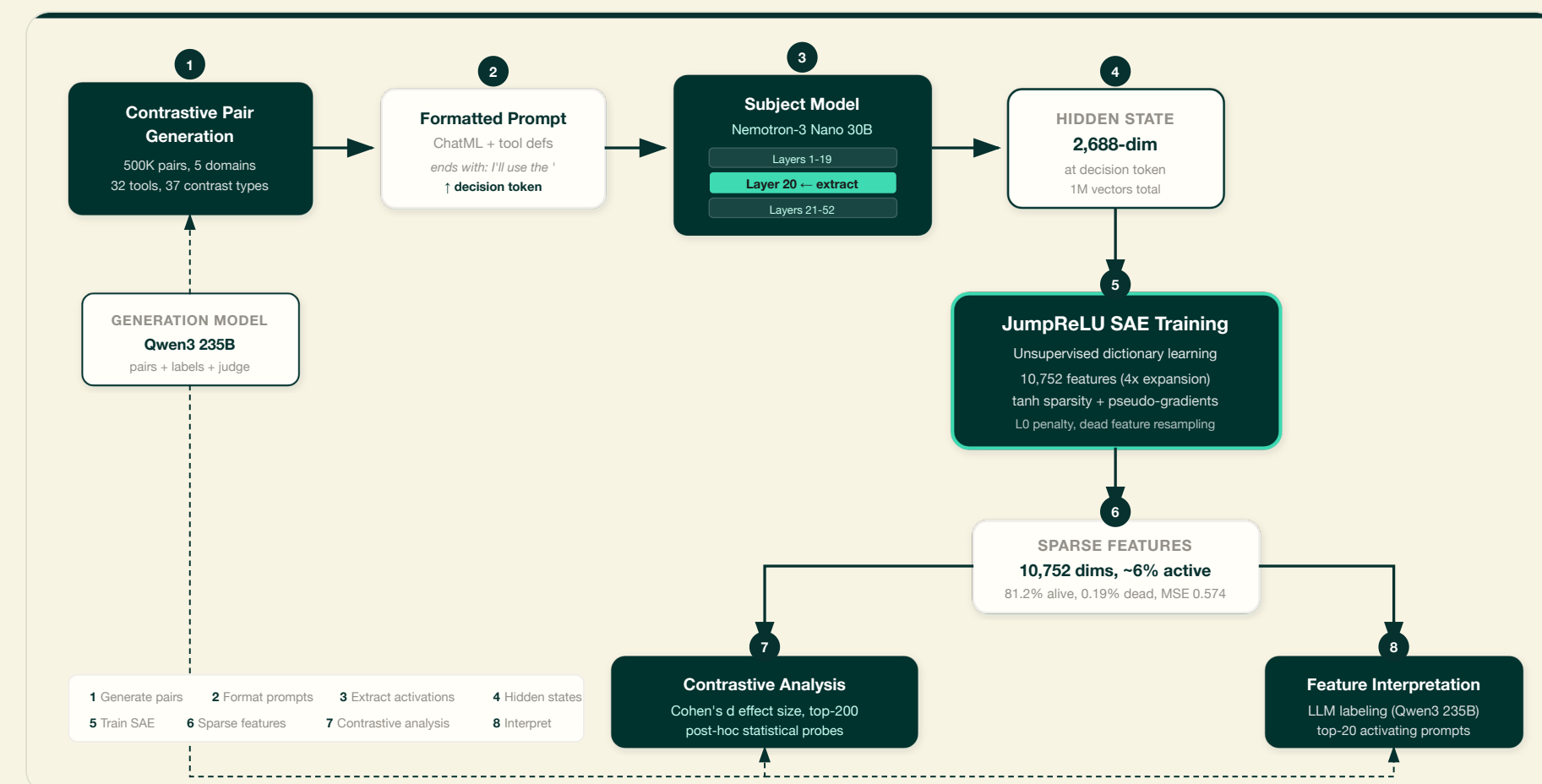


Fig 1. The SAE is trained **unsupervised** on raw activations. Contrastive pairs enter only at the post-hoc probe (Cohen's d) — never as a training signal.

## 03 Pick the Right Layer — Not the MoE Ones

Nemotron-3-Nano-30B is a **52-layer** model. A single SAE cannot decompose its MoE layers — reconstruction MSE explodes by **>500×** at the sparse-routed expert layers.

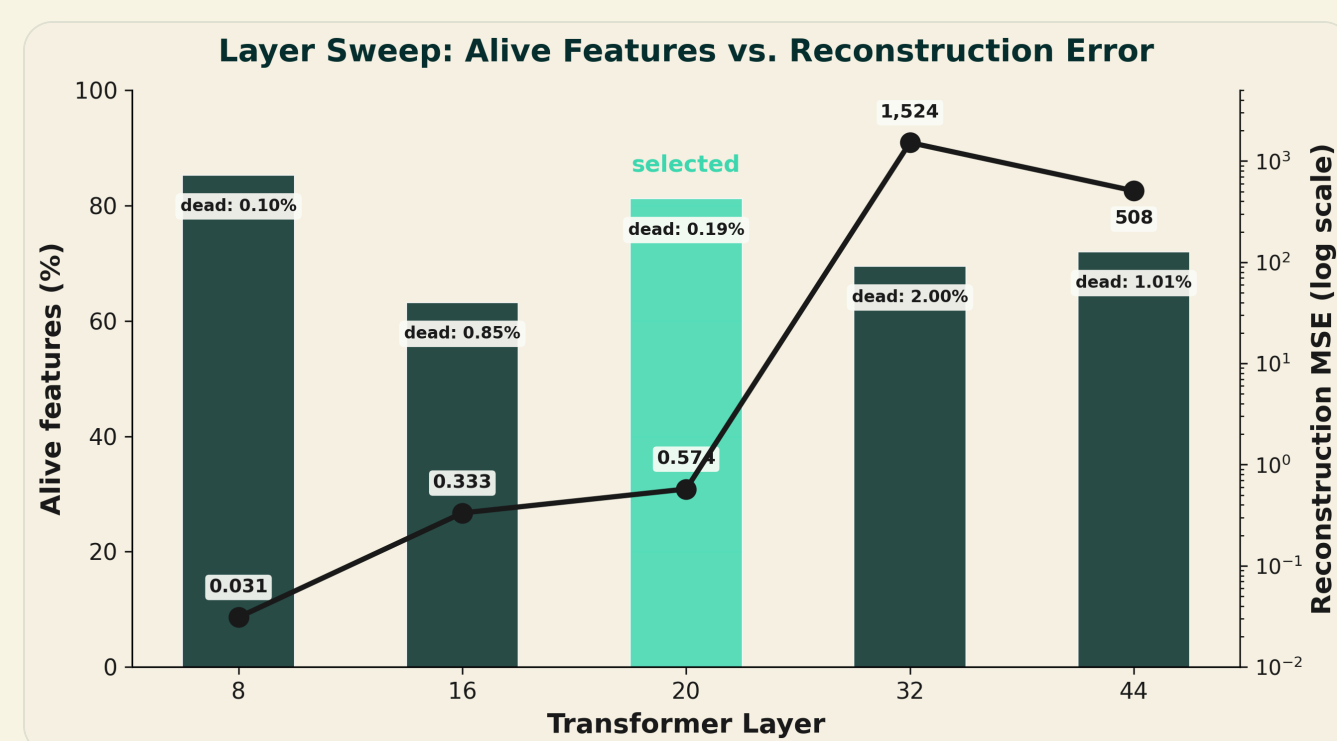
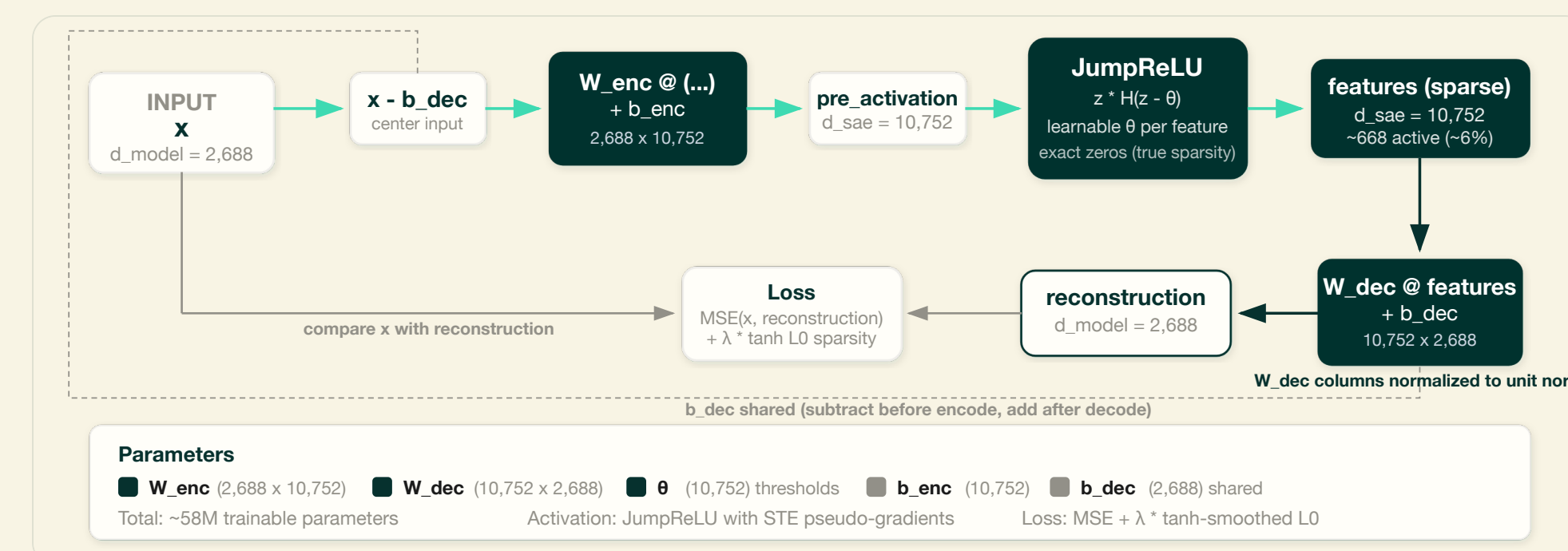


Fig 2. Bars = alive features; line = reconstruction MSE (log). At the MoE layers MSE jumps to **1,524** (L32) and **508** (L44) — vs. **0.574** at layer 20. Layer 20 sits just before the MoE stack with **81.2% alive features**.

## 04 JumpReLU Sparse Autoencoder

An overcomplete dictionary that decomposes hidden states into **monosemantic** features. JumpReLU thresholds  $\theta_i$  produce exact zeros — true sparsity — trained with a smooth tanh surrogate.



|                        |                    |
|------------------------|--------------------|
| Dictionary size M      | 10,752 (4× hidden) |
| Activations trained on | 1,000,000          |
| Reconstruction MSE     | 0.574              |
| Alive features         | 81.2%              |
| Active per input       | 668 ± 26 (6.2%)    |

Fig 3. Encode → JumpReLU (exact zeros) → decode, with shared  $b_{dec}$  and unit-norm  $W_{dec}$ . ~58M trainable parameters; loss = MSE + tanh-smoothed L0.

## 05 Inside a Feature — Emergent, Unsupervised Concepts

The SAE decomposes Layer-20 activations into **10,752 features**. Without contrastive supervision, named concepts emerge from raw activations alone:

- “internal knowledge retrieval”
- “data modification intent”
- “query complexity”

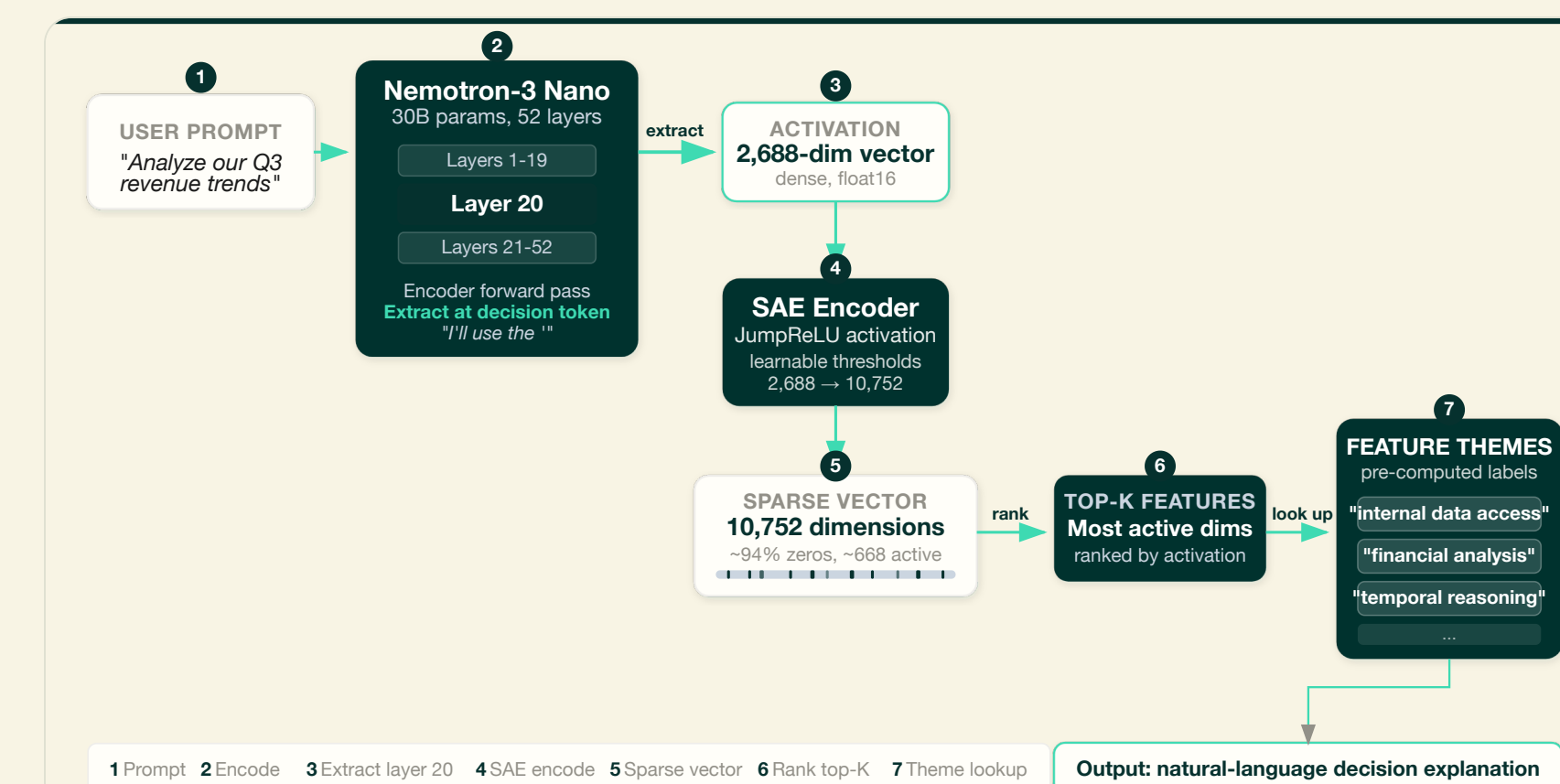
Each feature carries an **LLM-generated label**, a token-level fuzzing score (aggregate  $0.912 \pm 0.008$ ,  $p < 10^{-4}$ ), and its row in the contrastive-ablation table.

## Scope & Limitations

- One model, one layer, one decision-token convention.
- Top-10 ablation is conservative; distributed decisions need larger sets.
- Labels rely on Qwen3-VL-235B — its errors propagate to rationales.

## 06 Strong Focus on Production

Every tool call ships with a rationale. At inference, the top-K active features at the decision token map to pre-computed labels — **no extra LLM call**.



✓ human-readable explanation shipped with every tool call

Fig 4. Top-K active features map to pre-computed theme labels at the decision token — a debugging surface for deployed agents and a foundation for anomaly detection.

## 07 Feature Label Quality

# 84.3 %

Excellent (> 0.8)

Good 9.4 %  
Poor 6.2 %Combined score  $0.912 \pm 0.008$ 

Fig 5. 402 features · 4,422 A/B examples; combined score weights token-level 0.7 vs. prompt-level 0.3 ( $p < 10^{-4}$  vs. 0.5). Token-level evaluation catches labels right for the wrong reasons.

## RAI Responsible AI Motivation

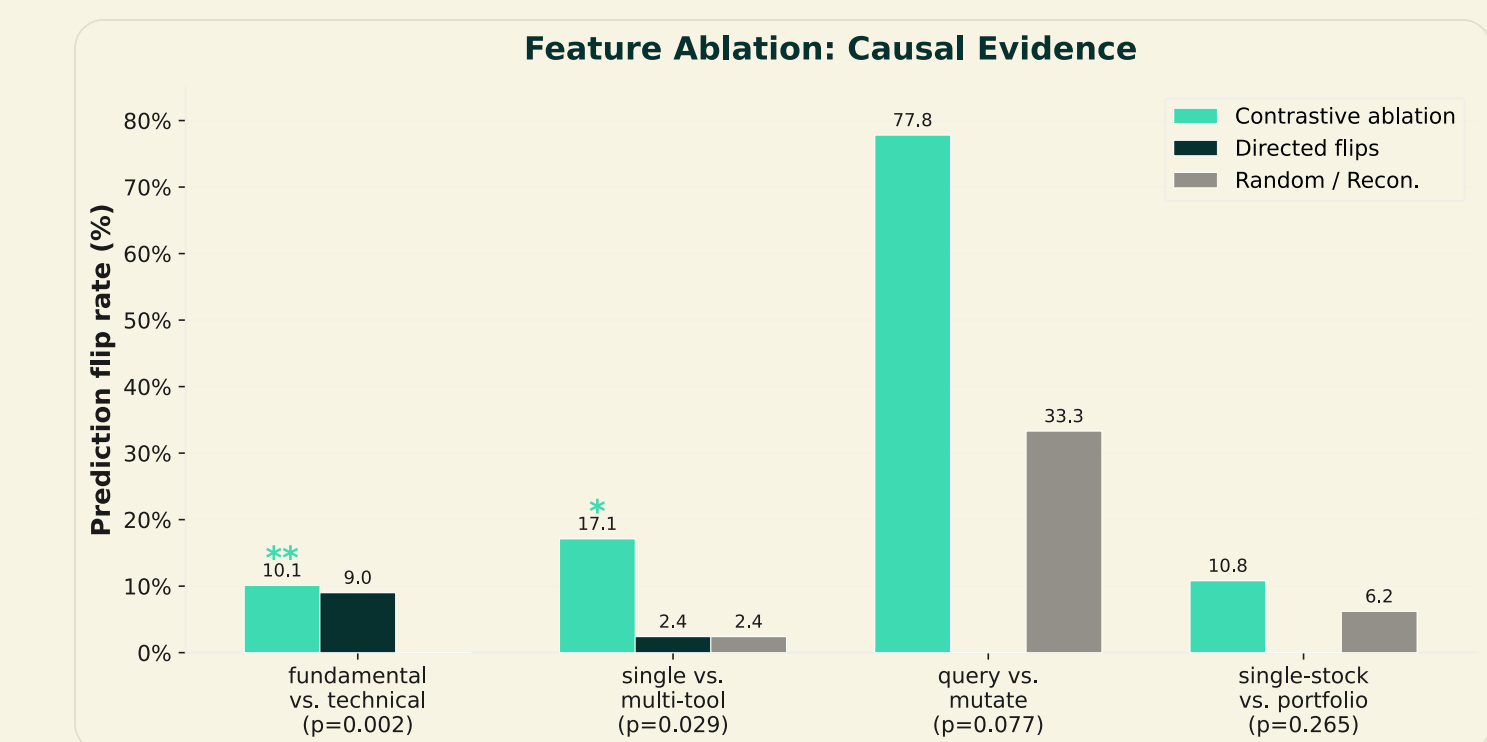
Transparency of agent decisions underpins trust as AI regulation expands. We ground this work in the **RAFT framework** [Gandhi 2025] — a lightweight value-criteria-indicator approach to Responsible AI.

- R** Reliable & Secure  
Consistent across the lifecycle; data and models secure, privacy-enhancing.
- F** Fair & Human-Centric  
Minimizes bias; supports human determination and choice.
- A** Accountable & Governed  
Lifecycle ownership documented for oversight and control.
- T** Transparent & Explainable  
AI use disclosed to users; inputs and outputs explained.

This work adds a mechanistic-interpretability indicator under the Transparent value.

## 08 Causal Ablation

Prediction-flip rate by contrast type — significance shown per group.



- Works: fundamental vs. technical — 10.1% flip, controls 0% ( $p = 0.002$ ).
- Works: single vs. multi-tool — 17.1% vs. 2.4% baseline ( $n = 41$ ,  $p = 0.029$ ).
- Null: 9 of 23 contrast types show zero flip — encoded distributively, not in a sparse circuit.

Why it's real: random-ablation flip equals SAE round-trip distortion across all 23 types. The contrastive flip sits **above this noise floor** — from the specific features removed, not encode/decode artefacts.

## 09 CATE Scaling — Distributed Causal Encoding

Targeted activation patching across expanding feature sets (Top 10/20/30) measures the **Conditional Average Treatment Effect**. The staircase rise signals abstract tool parameters governed by **distributed causal circuits**, not localized bottlenecks.

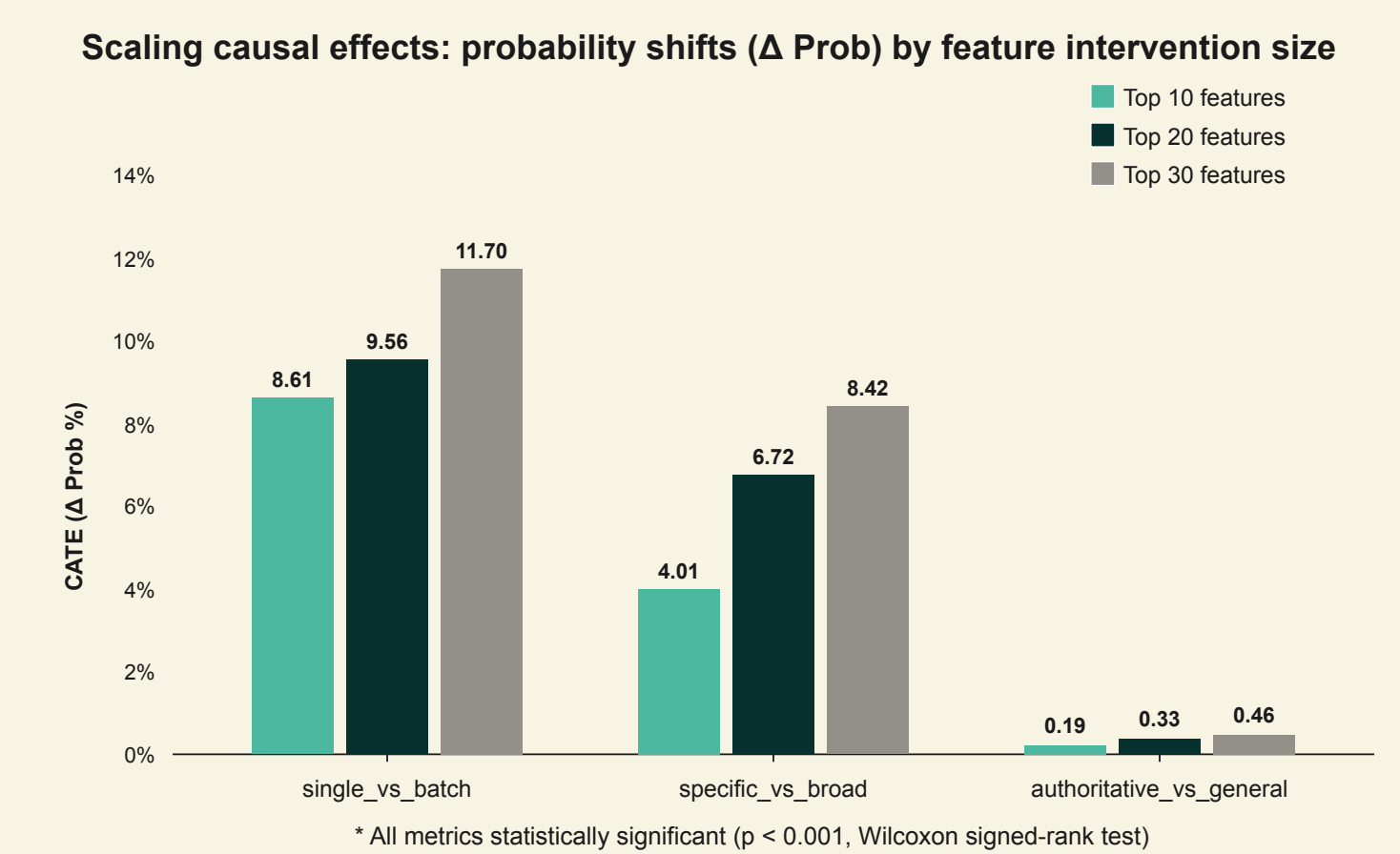


Fig 6. CATE ( $\Delta$  Prob %) by contrast type across Top-10 / 20 / 30 feature interventions; 500 prompts. The staircase rise with intervention size signals distributed causal encoding. All trajectories clear the Wilcoxon signed-rank test at  $p < 0.001$ .

## Code & Paper

[github.com/dataiku/kiji-inspector](https://github.com/dataiku/kiji-inspector)

Full paper, trained SAE weights, feature dashboard, and reproduction scripts.